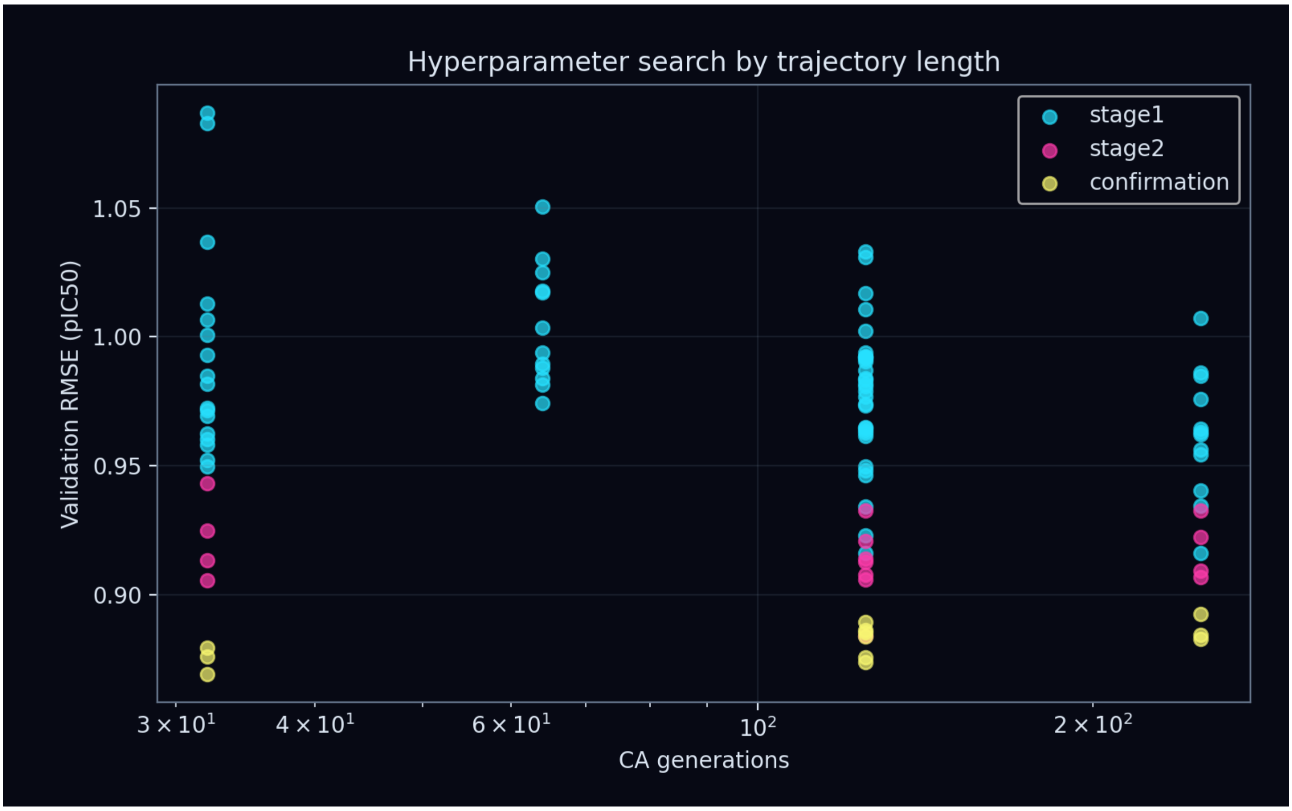
Strange Matter Engine

Production study: gray_scott

Direct-inhibition pIC50 | grouped validation | blinded set excluded

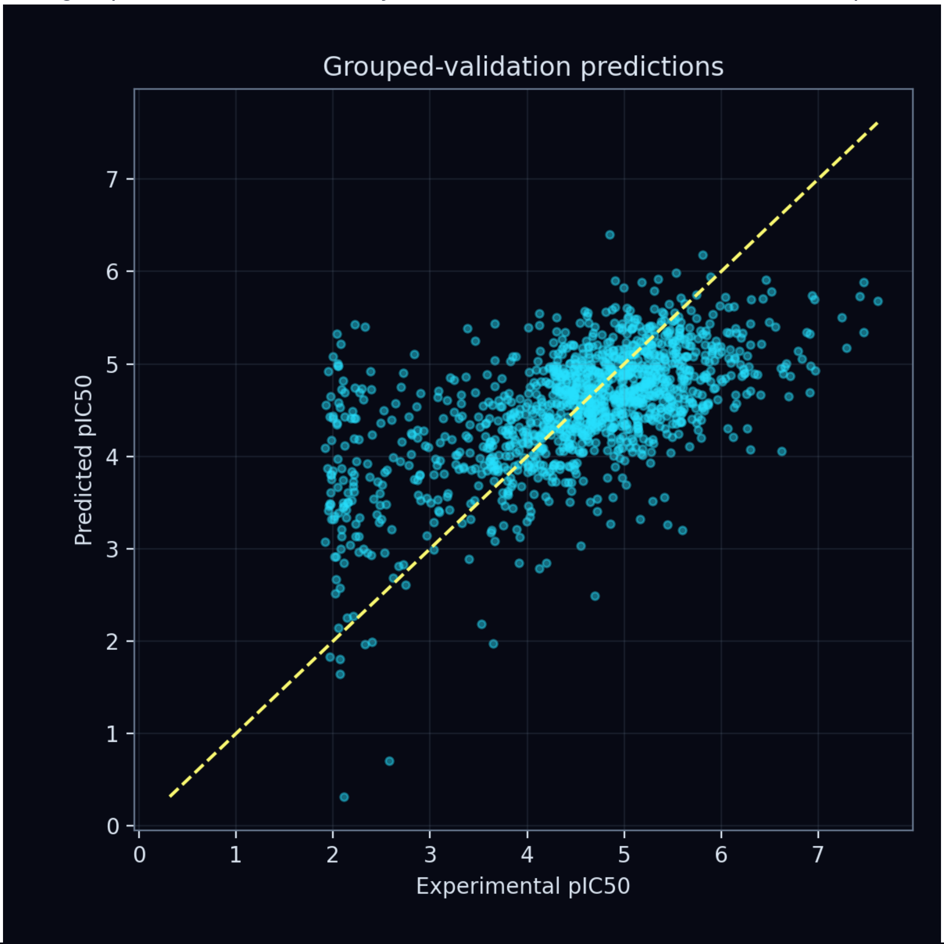
Quantity	Result
Validation RMSE	0.8651 pIC50
Fit RMSE	0.8087 pIC50
Generations	32
Dynamical channels	24
Atom feature profile	periodic_electronic
CA learning rate	0.003
Ridge penalty	0.01
CA L2	1e-05
Gradient clip	0.5
Confirmation mean	0.8749
Confirmation seed SD	0.0043

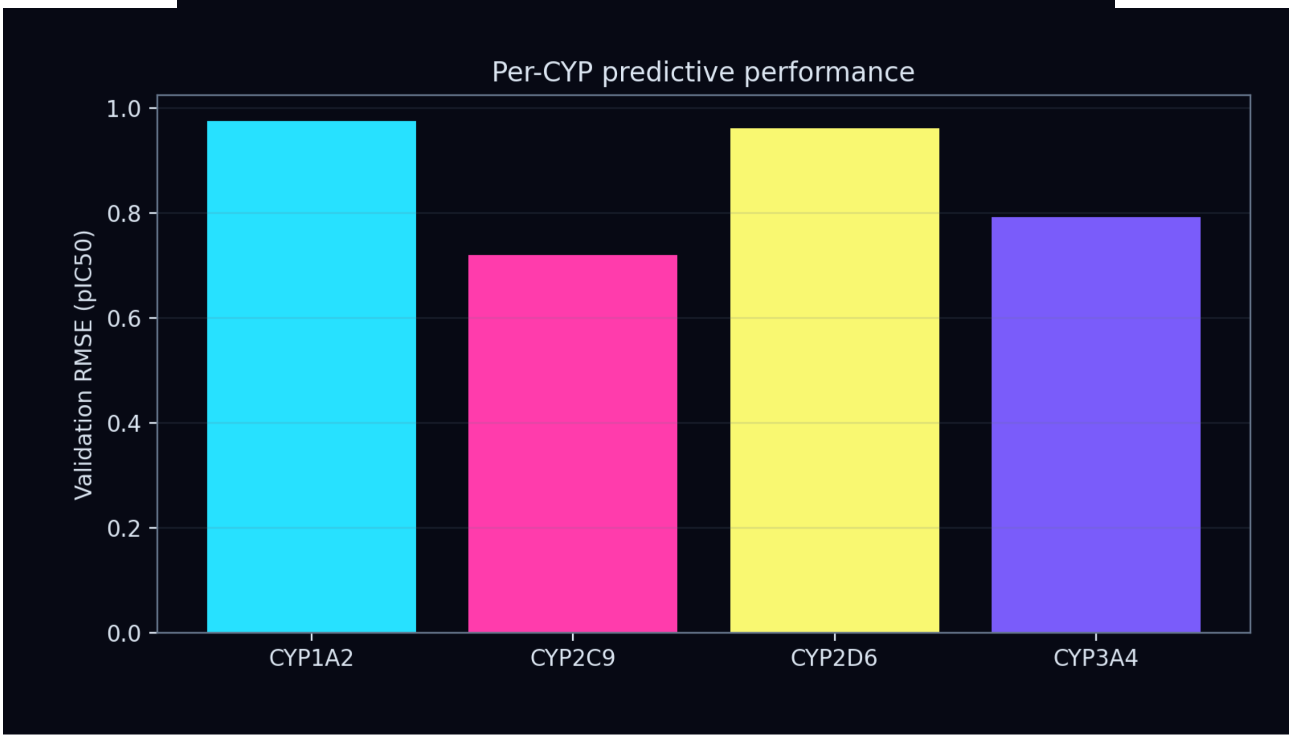
Enhanced CA controls: update scale 0.4, initial-state scale 1.5, support fraction 0.75, bond temperature 0.5. Rule dynamics: A=0.5, B=0.2, C=0.2, D=0.3.



Predictive validation

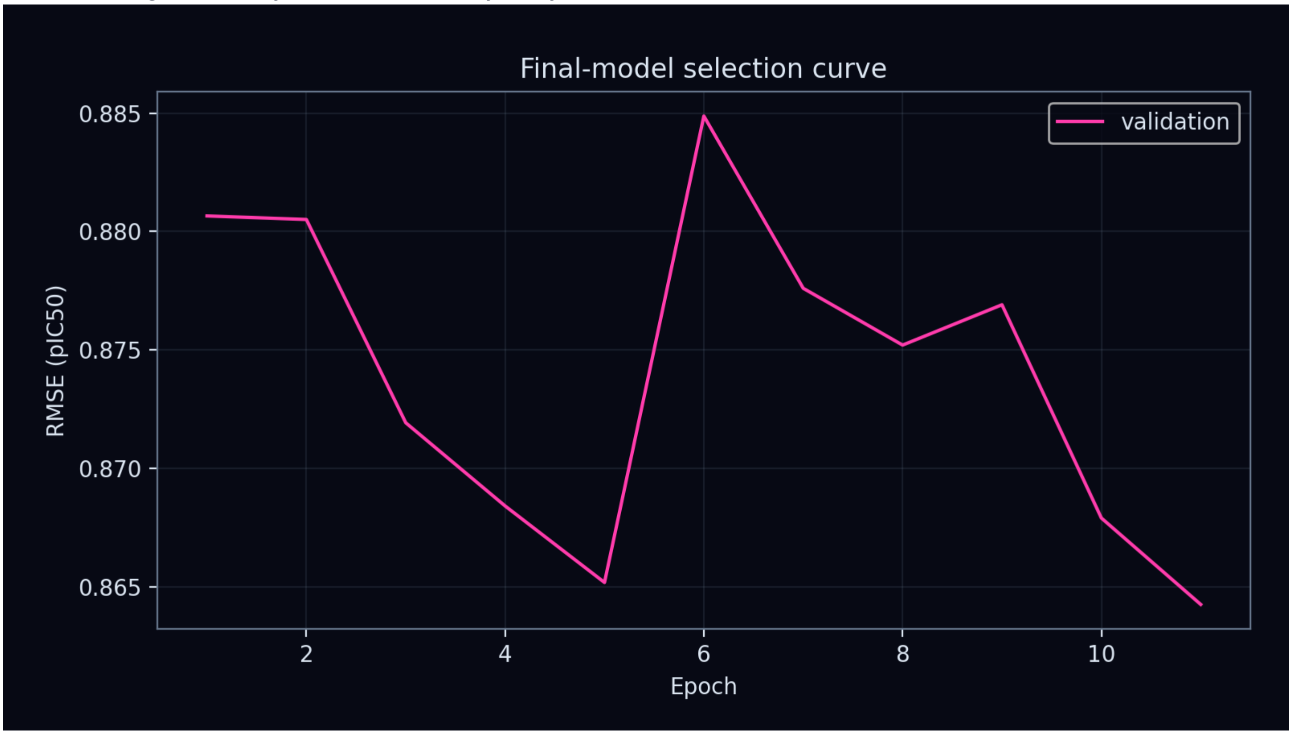
Model selection used grouped-validation RMSE. Dynamical-interest scores did not influence promotion.





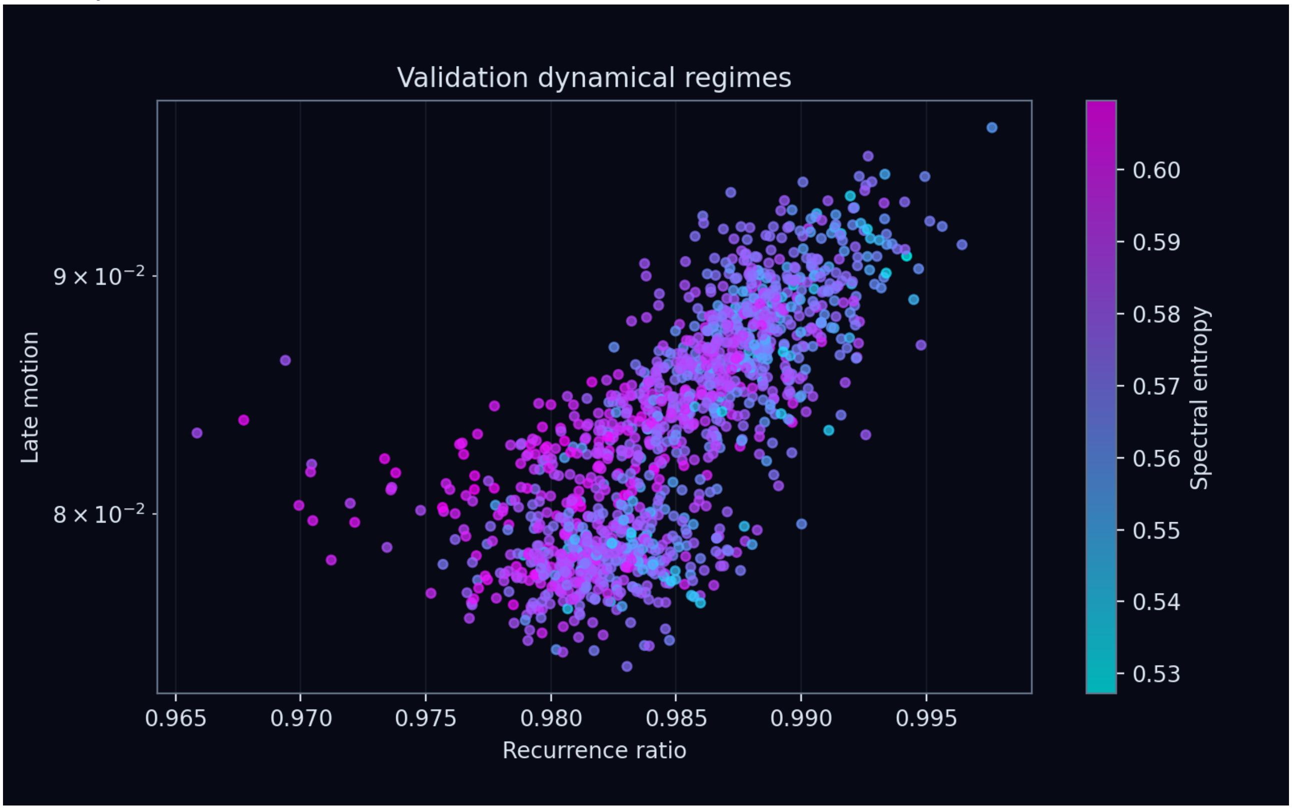
Optimization

Ridge coefficients and the unpenalized intercept were solved from permitted fitting fingerprints. Query error differentiated through the solve into the graph CA; Adam updated only CA parameters. Bond-conditioned messages, cosine learning-rate decay, and multitime trajectory statistics were shared across all ten rules.



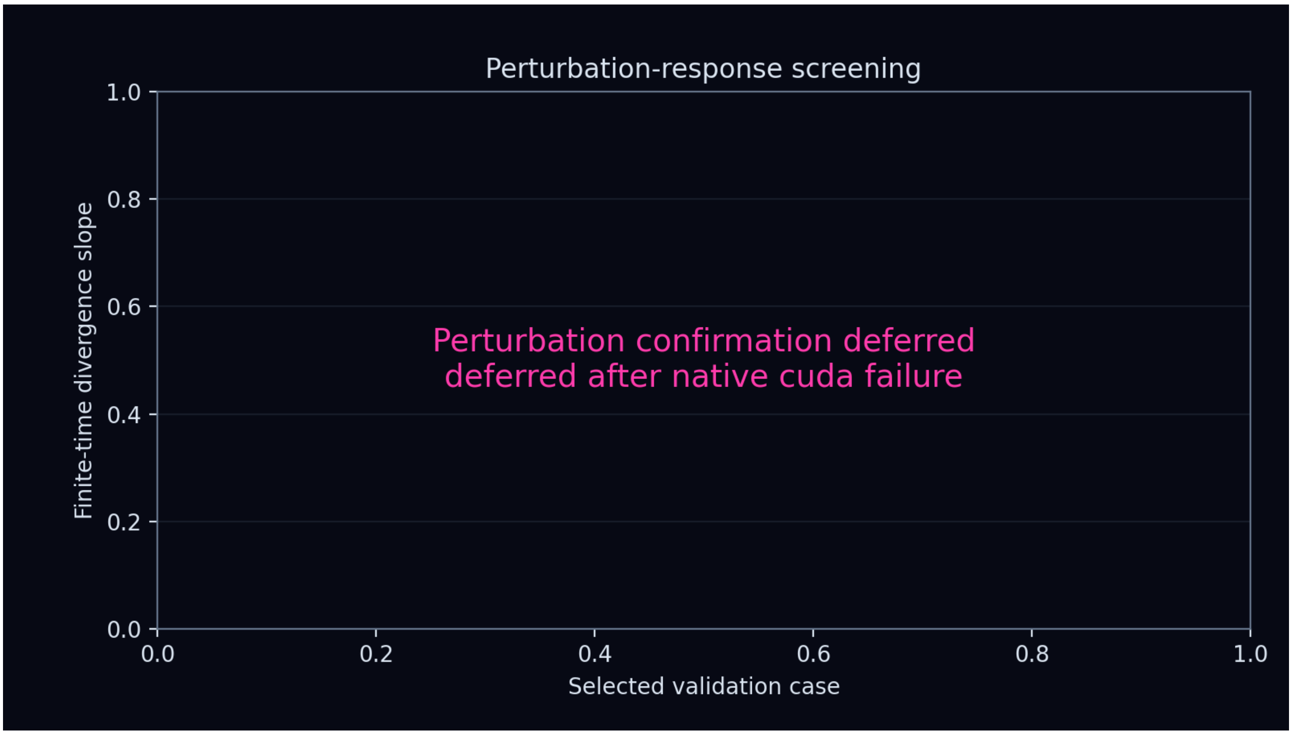
Emergent dynamics

The validation trajectories were screened for convergence, recurrence, persistent motion, spectral complexity, and perturbation sensitivity. Labels ending in candidate require longer, renormalized and seed-repeated confirmation before any attractor or chaos claim.



Permutation and perturbation screening

This figure tests whether small state perturbations separate or contract over the observed finite trajectory. It is placed on a dedicated page to preserve its axes, labels, and complete plotting area.



Scientific limitations

This report describes one transition-rule study under the declared grouped split. Hyperparameter selection and early stopping used labelled training data only. The blinded challenge set was not loaded or predicted. Finite-time positive slopes are screening signals rather than proof of a strange attractor or sustained chaos.